

SARASOTA BRADENTON INTL, FL, USA

WMO: 722115

Lat: 27.401N

Lon: 82.559W

Elev: 28

StdP: 14.68

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 12871

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 40.1	(c) 44.5	(d) 25.9	(e) 19.9	(f) 50.1	(g) 30.3	(h) 24.5	(i) 51.0	(j) 24.9	(k) 68.7	(l) 22.5	(m) 66.8	(n) 5.6	(o) 40	(p) 0.386

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 14.0	(c) 92.4	(d) 78.1	(e) 91.1	(f) 78.0	(g) 90.2	(h) 78.0	(i) 82.4	(j) 88.2	(k) 81.2	(l) 87.1	(m) 80.2	(n) 86.7	(o) 9.6	(p) 270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
81.1	161.8	86.3	79.4	152.6	84.6	78.6	148.6	84.3	46.2	88.4	44.7	86.7	43.5	86.4	87.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
20.7	18.3	16.4	33.1	96.0	3.9	2.3	30.3	97.7	28.0	99.0	25.8	100.3	23.0	102.0		
			29.7	82.7	4.3	1.8	26.6	84.0	24.1	85.1	21.7	86.1	18.6	87.5		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	73.6	61.5	64.3	67.5	72.2	77.4	81.3	82.7	82.8	81.6	76.9	69.6	65.1
	DBStd	9.10	7.86	7.31	6.24	4.39	3.33	2.44	2.12	1.98	2.35	4.81	5.91	7.65
	HDD50	17	10	3	0	0	0	0	0	0	0	0	0	4
	HDD65	434	159	93	50	6	0	0	0	0	0	3	27	96
	CDD50	8643	367	403	544	667	851	938	1013	1018	947	834	588	473
	CDD65	3585	51	73	128	223	386	488	548	553	497	372	165	101
	CDH74	33127	198	330	688	1568	3579	5110	6125	6126	4898	3080	965	460
	CDH80	11959	12	35	82	304	1166	2080	2612	2605	1899	960	161	43
(14) Wind	WSAvg	7.9	8.1	8.4	9.1	8.9	8.4	7.6	6.8	6.8	7.4	7.9	7.8	7.9
(15) Precipitation	PrecAvg	57.1	3.0	3.0	3.6	3.1	2.7	7.6	8.7	9.1	8.5	3.7	2.1	2.3
	PrecMax	76.0	7.9	7.9	10.1	9.6	7.8	21.1	14.6	16.5	14.3	7.8	8.7	11.8
	PrecMin	43.3	0.4	0.7	1.0	1.1	0.1	2.0	4.4	4.5	3.1	1.0	0.3	0.5
	PrecStd	7.9	1.9	1.7	2.5	1.8	2.0	3.9	2.4	3.3	3.1	1.9	1.7	2.7
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	81.8	83.8	84.9	88.0	91.5	93.4	94.1	94.5	93.1	90.8	86.1	83.2
		MCWB	71.0	71.6	68.9	71.0	72.7	77.6	78.7	78.8	77.5	76.5	73.8	72.7
	2%	DB	78.9	80.8	82.1	85.1	89.2	91.1	91.5	92.1	91.2	88.7	83.8	81.3
		MCWB	69.4	70.8	69.8	71.5	73.2	77.7	78.9	79.1	77.3	75.9	73.1	71.5
	5%	DB	76.2	78.4	80.6	83.4	87.6	89.9	90.4	90.8	90.1	87.2	81.8	79.1
		MCWB	67.7	69.6	69.6	71.4	73.5	77.9	78.8	79.0	77.5	75.5	71.6	70.4
	10%	DB	73.3	75.3	78.5	81.6	85.7	88.4	89.6	89.9	88.4	85.1	80.0	76.7
		MCWB	67.0	67.6	68.6	70.9	74.0	77.5	78.7	78.8	77.4	74.5	70.3	69.6
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	74.1	74.8	76.0	77.4	80.2	83.3	84.0	83.8	83.6	81.2	77.6	75.9
		MCDB	78.9	80.1	80.8	82.2	83.8	89.4	89.4	90.6	88.3	85.3	82.1	80.0
	2%	WB	71.6	73.0	74.0	75.5	78.2	81.4	82.1	81.8	81.1	79.2	76.1	74.1
		MCDB	75.5	78.2	78.9	81.5	83.9	87.2	87.9	88.5	86.0	84.1	81.1	78.4
	5%	WB	69.9	71.3	72.5	74.3	77.0	79.9	80.7	80.6	79.7	77.9	74.5	72.7
		MCDB	74.2	76.0	77.0	80.4	83.8	86.8	87.2	87.6	86.2	84.0	79.3	76.9
	10%	WB	68.2	69.6	70.8	73.0	75.9	78.4	79.6	79.7	78.5	76.7	72.6	70.8
		MCDB	72.4	74.3	75.4	79.2	82.8	85.9	86.8	87.1	85.4	82.9	77.2	75.0
(35) Mean Daily Temperature Range	5% DB	MDBR	18.3	18.0	18.0	17.5	16.6	14.2	13.7	14.0	14.0	15.9	17.7	17.4
		MCDBR	18.8	18.4	19.2	18.8	18.9	16.1	15.1	15.5	15.6	16.0	17.6	17.2
		MCWBR	10.5	9.9	9.1	8.5	7.4	6.5	6.2	6.1	5.9	6.3	8.7	9.1
	5% WB	MCDBR	15.9	15.5	14.4	15.3	14.4	13.5	13.6	14.0	13.7	13.4	14.3	14.7
		MCWBR	10.9	9.7	9.1	8.8	7.2	6.6	6.6	6.7	6.4	6.4	8.7	9.8
(40) Clear-Sky Solar Irradiance	taub		0.344	0.358	0.369	0.393	0.422	0.445	0.467	0.447	0.422	0.391	0.355	0.347
	taud		2.530	2.500	2.454	2.355	2.331	2.348	2.288	2.361	2.395	2.465	2.509	2.548
	Ebn at Noon		287	291	293	287	277	269	263	268	273	276	280	281
	Edh at Noon		29	32	35	40	41	40	43	39	37	33	29	27
(44) All-Sky Solar Radiation	RadAvg		1147	1394	1749	2089	2210	2005	1912	1832	1658	1533	1253	1064
	RadStd		85	89	138	91	117	165	114	112	107	91	64	86

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	-7	-130	+506	+383
(47) Regional (30 neighbors)	+0.77	+1.55	N/A	N/A	N/A	N/A	-5	-77	+304	+211

Nomenclature: See separate page